Algorithm Analysis And Data Structures

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Example for Running Time Calculation

Example:

$$\sum_{i=1}^{N} \sum_{j=i}^{N} \sum_{k=i}^{j} 1 = ?$$

$$\sum_{i=1}^{N}\sum_{j=i}^{N}\sum_{k=i}^{j}1=?$$

```
int sum(unsigned int n )
{ int sum;

  sum = 0;                                   1
  for( int i=1; i<=n; i++ )
    for(int j=i; j<=n; j++ )
      for(int k = i; k<=j; k++ )
        sum += 1;
  return(sum);                               1
}
```

sum += 1; → 2

inner loop: 2n+2, n × 2 → 2n, total 4n+2

middle loop: 2n+2, n × (4n+2) → $4n^2$+2n, total $4n^2$+4n+2

outer loop: 2n+2, n × ($4n^2$+4n+2) → $4n^3$+$4n^2$+2n

T($4n^3$+$4n^2$+4n+4+1+1) = O (n^3)

$O(N^3)$

Algorithm 1

$$\sum_{k=i}^{j} 1 = j - i + 1$$

```
int sum(unsigned int n )
{ int sum;

  sum = 0;
  for( int i=1; i<=n; i++ )
   for(int j=i; j<=n; j++ )
     sum += j-i+1;

  return(sum );
}
```

1

2n+2

2n+2 n $6n^2$+2n $6n^2$+4n+2

4 n 4n 6n+2

1

T($6n^2$+6n+2+1+1) = O (n^2)

$O(N^2)$

Algorithm 2

$$\sum_{i=1}^{N} \sum_{j=i}^{N} \sum_{k=i}^{j} 1 = ?$$

$$\sum_{k=i}^{j} 1 = j - i + 1$$

$$\sum_{j=i}^{N} j - i + 1 = \sum_{p=1}^{N-i+1} p = \frac{(N-i+1).(N-i+2)}{2} \quad (p = j - i + 1)$$

Divide&Conquer Algorithm

First Half Center Second Half

---------------------- ---------------------

Left

Right

left_sum

right__sum

left_sum + right_sum

```
int sum(int left, int right )
{       int left_sum, right_sum;
        int center, sum;

        center = (left + right )/2;

        left_sum = 0;
        for(int i=center; i>=left; i-- )
                left_sum += (n-i+1)*(n-i+2)/2;

        center = center+1

        right_sum = 0;
        for(int i=center; i<=right; i++ )
                right_sum += (n-i+1)*(n-i+2)/2;

        sum = left_sum + right_sum;
        return(sum);
}
```

Algorithm 3 - Divide&Conquer Algorithm

O(NlogN)

$$\sum_{j=i}^{N} j-i+1=\sum_{p=1}^{N-i+1} p=\frac{(N-i+1).(N-i+2)}{2} \quad (p=j-i+1)$$

```
int sum(unsigned int n )
{         int sum;

          sum = 0;
          for( i=1; i<=n; i++ )
                    sum += (n-i+1)*(n-i+2)/2;

          return(sum );
}
```

Algorithm 4

O(N)

$$\sum_{i=1}^{N} \sum_{j=i}^{N} \sum_{k=i}^{j} 1 = ?$$

$$\sum_{i=1}^{N} \frac{(N-i+1).(N-i+2)}{2}$$

$$= \frac{1}{2}\sum_{i=1}^{N} i^2 - \left(N + \frac{3}{2}\right)\sum_{i=1}^{N} i + \frac{1}{2}\left(N^2 + 3N + 2\right)\sum_{i=1}^{N} 1$$

$$= \frac{1}{2}\frac{N(N+1)(2N+1)}{6} - \left(N + \frac{3}{2}\right)\frac{N(N+1)}{2} + \frac{N^2 + 3N + 2}{2} N$$

$$= \frac{N^3 + 3N^2 + 2N}{6}$$

```
int sum(unsigned int n )
{         int sum;

          sum = 0;
          sum += (n*n*n + 3*n*n + 2*n) / 6;

          return(sum );
}
```

$O(12) = O(c)$

Algorithm 5

Questions and Answers

Thank you

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